Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LEARNER GUIDE

For

Financial Analysis for Small and Medium Enterprises

TGS Ref No: TGS-2026064860

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 11

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
10.0	1 June 2026	Legacy learner guide aligned to the v10 master trainer slides.	Han Leong
11	10 August 2026	Full redesign: regenerated from the single-source content pipeline with six step-by-step hands-on activities (cash flow statement, ratio analysis, trend analysis, payback, NPV/PI, solvency), revision pointers and a glossary.	Dr Alfred Ang

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Introduction

This Learner Guide accompanies the WSQ course Financial Analysis for Small and Medium Enterprises (TGS-2026064860), conducted by Tertiary Infotech Academy Pte Ltd. It provides the key concepts for each of the three topics and detailed step-by-step instructions for all six hands-on activities. Each activity uses a worksheet downloadable from the course LMS (https://lms-tms.tertiaryinfotech.com) and is completed in Microsoft Excel or Google Sheets.

Use this guide alongside the course slides. The final assessment is open book: you may refer to the slides, this Learner Guide and any approved materials, so keep your completed activity worksheets — they are your best revision notes.

Course Learning Outcomes

- LO1: Understand the financial statements such as balance sheet, income, and cash flow statements.
- LO2: Evaluate organization's financial performance from the trend of financial ratios.
- LO3: Analyze financial statements and prepare the organization's position.

Skills Framework (TSC)

TSC Title: Financial Analysis · TSC Code: ACC-MAC-5004-1.1

- K1: Statement of financial position
- K2: Balance sheet
- K3: Income and cash flow statements
- K4: Statement of changes in equity
- K5: Financial statement analysis techniques
- A1: Identify trends by comparing ratios across multiple time periods and statement types
- A2: Prepare and interpret performance and position of an organisation using financial statements

Before You Start — Setup

What you need

- A laptop with Microsoft Excel (2016 or later) or a Google account for Google Sheets.
- Access to the course LMS at https://lms-tms.tertiaryinfotech.com — log in with your registered email (an OTP is sent to you).
- The activity worksheets, downloaded from your course page on the LMS.
- A calculator (or the spreadsheet itself) for the ratio and discounting computations.

Conventions used in every activity

- Figures in the Topic 1 activity are in $ million; Topic 2 and 3 activities use dollars.
- Workings shown in a box are the expected calculation or Excel formula for that step.
- Each activity ends with a 'Test it' check — compare your result against it before moving on.
- Model answers are discussed in class after each activity.

Topic 01 — Understanding Financial Statements (K1 · K2 · K3)

Overview of Finance and Chart of Accounts · Balance Sheet Statement · Profit and Loss (P&L) Statement · Cash Flow Statement

Key concepts

- Finance & accounting — Why finance matters to every business decision — the language of the boardroom.
- Chart of accounts — The general-ledger account structure that every transaction is recorded against.
- Balance sheet — Assets = Liabilities + Equity — the statement of financial position at a point in time.
- Income statement — Revenue − Expenses = Profit — financial performance over an accounting period.
- Cash flow statement — Operating, investing and financing cash movements — where the cash actually went.
- Double-entry system — Debits equal credits, so the accounting equation always balances.

Activity 1 — Prepare a Cash Flow Statement

Maps to: LO1 — understand the balance sheet, income and cash flow statements (K1, K2, K3).

Goal: You are given the Balance Sheet (2018 and 2019) and the Income Statement (2019) of a company, all figures in $ million. Working in Excel or on paper, derive the statement of cash flows — operating, investing and financing — and reconcile it to the change in cash.

What you'll produce

A complete statement of cash flows (operating / investing / financing) derived from the balance sheet and income statement. (Tools: Microsoft Excel (or Google Sheets), activity worksheet from the LMS.)

Data provided

- Balance Sheet 2019: Fixed Assets 100 · Receivables 50 · Inventory 30 · Cash 20 · Total Assets 200 | Equity 80 · Loans 50 · Payables 40 · Provisions 30 · Total 200

- Balance Sheet 2018: Fixed Assets 90 · Receivables 50 · Inventory 30 · Cash 20 · Total Assets 190 | Equity 80 · Loans 55 · Payables 35 · Provisions 20 · Total 190
- Income Statement 2019: Revenue 50 · COGS 10 · Operational costs 20 · Depreciation 5 · Interest 3 · Tax 2 · Net Income 10

Step-by-step

1. Download the activity worksheet 'Cash Flow Statement' from the LMS (lms-tms.tertiaryinfotech.com) and open it in Excel.
2. Compute EBITDA from the income statement: Revenue − COGS − Operational costs = 50 − 10 − 20 = 20.

```
Working:  EBITDA = 20
```

3. Compute Net Working Capital movement: NWC = Payables − Receivables − Inventory = 40 − 50 − 30 = −40 (2019) vs 35 − 50 − 30 = −45 (2018); the change is −5.

```
Working:  ΔNWC = −5
```

4. Build the Operating Cash Flow block: EBITDA 20, Tax −6, Change in NWC −5, Change in Provisions +10 → Operating Cash Flow = 18.

```
Working:  OCF = 20 − 6 − 5 + 10 = 18
```

5. Build the Investing Cash Flow block: Change in Fixed Assets −10, Depreciation 2019 −5, Change in trade payables on capex +1 → Investing Cash Flow = −14.

```
Working:  ICF = −10 − 5 + 1 = −14
```

6. Build the Financing Cash Flow block: Change in financial debt −5 (loans 55 → 50), Interest cost −3, Change in earnings −6 → Financing Cash Flow = −14.

```
Working:  FCF = −5 − 3 − 6 = −14
```

7. Compute the Net Cash Flow and reconcile: 18 − 14 − 14 = −10.

```
Working:  Net Cash Flow = −10
```

Test it

Your three blocks total 18 (operating), −14 (investing) and −14 (financing), giving a net cash flow of −10 that reconciles with the balance-sheet movement.

Note: The full worksheet for this activity is available on the LMS, and a printable copy is in activities/activity-01 of the course repository.

Topic 02 — Analysing Financial Ratios (K4 · A1)

Ratios for Corporate Profitability · Ratios for Corporate Performance · Equity Changes Statement

Key concepts

- Liquidity ratios — Current, quick, cash and operating-cash-flow ratios — can the firm pay its short-term bills?
- Leverage ratios — Debt, debt-to-equity, interest coverage — how much risk is carried in the capital structure?
- Efficiency ratios — Inventory, receivables, payables and asset turnover — how hard are the assets working?
- Profitability ratios — Gross, operating, net margins, ROA, ROCE, ROE, EPS — is the business earning enough?
- Equity changes statement — Reconciles opening to closing equity — share capital, dividends, retained earnings.
- Trend analysis — Compare ratios across periods and statement types to spot direction and risk (A1).

Activity 2 — Ratio Analysis of Two Companies

Maps to: LO2 — evaluate financial performance from financial ratios (K4, A1).

Goal: You are a financial analyst evaluating two companies, X and Y, from their balance sheets and income statements. Compute the liquidity, profitability, turnover and solvency ratios for both companies and comment on which company performs better on each parameter.

What you'll produce

A completed ratio-analysis worksheet comparing Company X and Company Y across four ratio families, with a one-line verdict per ratio. (Tools: Microsoft Excel (or Google Sheets), activity worksheet from the LMS.)

Data provided

- Balance Sheet (X | Y): Cash 50,000 | 80,000 · Inventory 40,000 | 70,000 · AR 100,000 | 250,000 · Property 400,000 | 450,000 · Other FA 60,000 | 50,000 · Total 650,000 | 900,000
- Liabilities (X | Y): AP 100,000 | 150,000 · ST Loans 80,000 | 70,000 · LT Liabilities 300,000 | 500,000 · Capital 170,000 | 180,000
- Income Statement (X | Y): Sales 300,000 | 500,000 · COGS 140,000 | 200,000 · S&D 50,000 | 60,000 · EBIT 100,000 | 228,000 · Interest 10,000 | 18,000 · Tax 18,000 | 42,000 · Net Income 72,000 | 168,000

Step-by-step

8. Download the activity worksheet 'Ratio Analysis' from the LMS and open it in Excel.
9. Liquidity — Current Ratio = Current Assets / Current Liabilities. X: 190,000/180,000 = 1.06; Y: 400,000/220,000 = 1.82. Y is more liquid.

`Working:  X 1.06 · Y 1.82`

10. Liquidity — Quick Ratio = (Current Assets − Inventory) / Current Liabilities. X: 150,000/180,000 = 0.83; Y: 330,000/220,000 = 1.50.

`Working:  X 0.83 · Y 1.50`

11. Profitability — Operating Profit Ratio = EBIT / Sales. X: 100,000/300,000 = 33%; Y: 228,000/500,000 = 46%.

`Working:  X 33% · Y 46%`

12. Profitability — Net Profit Ratio = Net Income / Sales. X: 72,000/300,000 = 24%; Y: 168,000/500,000 = 34%.

`Working:  X 24% · Y 34%`

13. Profitability — Return on Capital Employed = EBIT / Capital Employed. X: 100,000/470,000 = 21%; Y: 228,000/680,000 = 34%.

`Working:  X 21% · Y 34%`

14. Turnover — Inventory Turnover = COGS / Average Inventory. X: 140,000/40,000 = 3.50; Y: 200,000/70,000 = 2.86. X turns stock faster.

`Working:  X 3.50 · Y 2.86`

15. Turnover — Receivables Turnover = Credit Sales / Average Receivables. X: 300,000/100,000 = 3.00; Y: 500,000/250,000 = 2.00.

`Working:  X 3.00 · Y 2.00`

16. Turnover — Payables Turnover (assume purchases = 50% of COGS). X: 70,000/100,000 = 0.70; Y: 100,000/150,000 = 0.67.

`Working:  X 0.70 · Y 0.67`

17. Solvency — Debt-to-Equity = Long-Term Debt / Equity. X: 300,000/170,000 = 1.76; Y: 500,000/180,000 = 2.78. Y is more leveraged.

`Working:  X 1.76 · Y 2.78`

18. Solvency — Financial Leverage = Total Assets / Equity. X: 650,000/170,000 = 3.82; Y: 900,000/180,000 = 5.00.

`Working:  X 3.82 · Y 5.00`

19. Write a one-line verdict per ratio family: Y wins on liquidity and profitability; X wins on turnover efficiency and carries less solvency risk.

Test it

Your worksheet shows all ten ratios for both companies and your verdicts match the model answers (e.g. Current Ratio X 1.06 vs Y 1.82; D/E X 1.76 vs Y 2.78).

Note: The full worksheet for this activity is available on the LMS, and a printable copy is in activities/activity-02 of the course repository.

Activity 3 — Trend Analysis over Four Years

Maps to: LO2 — identify trends by comparing ratios across time periods (K5, A1).

Goal: You are given four years (2018–2021) of a company's balance sheet and income statement. Compute year-on-year growth for every line item, derive the key ratios for each year, and present your observations on the company's direction.

What you'll produce

A completed trend-analysis template: YoY growth for every balance-sheet and income-statement line plus a four-year ratio table with your observations. (Tools: Microsoft Excel (or Google Sheets), activity template from the LMS.)

Data provided

- Sales 2018→2021: 180,000 → 198,000 → 217,800 → 239,580 (10% p.a.)
- Net Income 2018→2021: 53,600 → 64,601 → 76,872 → 90,548
- Total Assets 2018→2021: 330,000 → 324,800 → 329,983 → 339,581

Step-by-step

20. Download the activity template 'Trend Analysis' from the LMS and open it in Excel.
21. Fill the YoY Growth columns of the balance sheet: growth = (this year − last year) / last year. E.g. Inventory 2021 = (40,000 − 34,000)/34,000 = 17.65%.

```
Working:  =(C5-E5)/E5
```

22. Fill the YoY Growth columns of the income statement. Sales grow a steady 10% each year while COGS grows only 5% — so Gross Profit growth accelerates (14.47% → 13.78%).
23. Compute the liquidity ratios per year: Current Ratio falls from 2.13 (2018) to 1.64 (2021); Acid Ratio falls from 1.47 to 1.22.

```
Working:  Current = CA/CL
```

24. Compute the efficiency ratios per year: Asset Turnover rises 0.55 → 0.72; Inventory Turnover rises 4.00 → 6.48; Receivables Turnover rises 3.00 → 3.29.
25. Compute the leverage ratios per year: Debt-to-Equity rises 0.56 → 0.75; Debt-to-Assets rises 0.36 → 0.43.
26. Compute the profitability ratios per year: Operating Margin rises 0.40 → 0.49; Net Profit Margin rises 0.30 → 0.38.
27. Write your observations: profitability and efficiency improve steadily; liquidity declines and leverage creeps up — flag the falling current ratio as the trend to watch.

Test it

Your ratio table matches the model answers (e.g. 2021 Current Ratio 1.64, Net Profit Margin 0.38) and your observations cover profitability, efficiency, liquidity and leverage trends.

Note: The full worksheet for this activity is available on the LMS, and a printable copy is in activities/activity-03 of the course repository.

Topic 03 — Planning & Budgeting using Financial Statements (K5 · A2)

Analyse Financial Statements · Financial Planning · Capital Budgeting

Key concepts

- Budgeting — Baseline, incremental, zero-based and hybrid budgets — planning income vs spending.
- Capital budgeting — Payback, discounted payback, NPV, PI and IRR — evaluating long-term investments.
- Time value of money — PV = FV / (1+i)^n — a dollar today is worth more than a dollar tomorrow.
- Forecast & variance — Budget vs actual vs forecast — favourable and adverse variances, thresholds.
- Financial health — Read the balance sheet, income and cash flow statements together, then ratio-check.
- Analysis methods — Ratio, horizontal (trend) and vertical analysis, plus industry benchmarking (A2).

Activity 4 — Payback Period & Discounted Payback Period

Maps to: LO3 — evaluate a capital investment using payback methods (K5, A2).

Goal: A project requires an initial outlay of $2,000 and returns $375 per year for 10 years. Compute the simple payback period, then repeat the exercise discounting each cash flow at 10% to find the discounted payback period.

What you'll produce

A payback worksheet showing cumulative cash flow per year, the simple payback point (~5.3 years) and the discounted payback point (~8 years). (Tools: Microsoft Excel (or Google Sheets), activity worksheet from the LMS.)

Data provided

- Cash flows: Year 0: −$2,000; Years 1–10: +$375 per year
- Discount rate: 10% for the discounted payback

Step-by-step

28. Download the activity worksheet 'Cash Flow Payback' from the LMS and open it in Excel.
29. Build the cumulative cash flow column: 375, 750, 1,125, 1,500, 1,875, 2,250 … the cumulative figure passes $2,000 during Year 6.

```
Working:  =SUM($B$2:B7)
```

30. Compute the simple payback period: 2,000 / 375 = 5.33 years (about 5 years 4 months).

```
Working:  Payback = 2000/375 = 5.33 yr
```

31. Add a discounted cash flow column: DCF = 375 / (1.10)^n → 340.91, 309.92, 281.74, 256.13, 232.85, 211.68, 192.43, 174.94 …

```
Working:  =375/(1.1)^A2
```

32. Build the cumulative discounted column: 340.91, 650.83, 932.57, 1,188.70, 1,421.55, 1,633.22, 1,825.66, 2,000.60 — it crosses $2,000 in Year 8.
33. State the discounted payback period: ≈ 8 years, and explain why discounting lengthens the payback.

```
Working:  Discounted payback ≈ 8.0 yr
```

Test it

Your cumulative discounted cash flow reaches $2,000.60 at Year 8 — the discounted payback (~8 years) is materially longer than the simple payback (5.33 years).

Note: The full worksheet for this activity is available on the LMS, and a printable copy is in activities/activity-04 of the course repository.

Activity 5 — Net Present Value & Profitability Index

Maps to: LO3 — evaluate a capital investment using NPV and PI (K5, A2).

Goal: Using the same project ($2,000 outlay, $375/year for 10 years, 10% required return), compute the Net Present Value and the Profitability Index, and decide whether to accept the project.

What you'll produce

An NPV/PI worksheet: PV of each inflow, NPV ≈ $304, PI ≈ 1.15, and an accept/reject decision. (Tools: Microsoft Excel (or Google Sheets), activity worksheet from the LMS.)

Data provided

- Cash flows: Year 0: −$2,000; Years 1–10: +$375 per year
- Required return: 10%

Step-by-step

34. Download the activity worksheet 'Discounted Cash Flow' from the LMS and open it in Excel.
35. Discount each inflow at 10%: 341, 310, 282, 256, 233, 212, 192, 175, 159, 145.

```
Working:  =375/(1.1)^A2
```

36. Sum the discounted inflows: PV of cash inflows = $2,304.

```
Working:  PV inflows = 2,304
```

37. Compute NPV = PV of inflows − initial outlay = 2,304 − 2,000 = $304 (or use =NPV(10%, range) − 2000).

```
Working:  NPV = 304
```

38. Compute the Profitability Index = PV of inflows / PV of outflows = 2,304 / 2,000 = 1.15.

```
Working:  PI = 1.15
```

39. Decide: NPV > 0 and PI > 1 → accept the project; note how the decision rule would flip if the discount rate rose.

Test it

Your worksheet shows NPV ≈ $304 and PI ≈ 1.15, and your decision is ACCEPT because NPV > 0 and PI > 1.

Note: The full worksheet for this activity is available on the LMS, and a printable copy is in activities/activity-05 of the course repository.

Activity 6 — Solvency & Financial Risk Analysis

Maps to: LO3 — compare companies' solvency risk from their statements (K5, A2).

Goal: Using the balance sheets and income statements of companies X and Y, compare the two companies in terms of solvency risk: debt-to-equity, financial leverage and interest coverage. Conclude which company is the riskier borrower.

What you'll produce

A solvency comparison table (D/E, leverage, interest coverage) for X and Y with a risk verdict. (Tools: Microsoft Excel (or Google Sheets), activity worksheet from the LMS.)

Data provided

- Balance Sheet (X | Y): Total Assets 650,000 | 900,000 · LT Debt 300,000 | 500,000 · Equity 170,000 | 180,000
- Income Statement (X | Y): EBIT 100,000 | 228,000 · Interest 10,000 | 18,000

Step-by-step

40. Download the activity worksheet 'Solvency Analysis' from the LMS and open it in Excel.
41. Compute Debt-to-Equity = Long-Term Debt / Equity. X: 300,000/170,000 = 1.76; Y: 500,000/180,000 = 2.78.

```
Working:  X 1.76 · Y 2.78
```

42. Compute Financial Leverage = Total Assets / Equity. X: 650,000/170,000 = 3.82; Y: 900,000/180,000 = 5.00.

```
Working:  X 3.82 · Y 5.00
```

43. Compute Interest Coverage = EBIT / Interest Expense. X: 100,000/10,000 = 10.0; Y: 228,000/18,000 = 12.7.

```
Working:  X 10.0 · Y 12.7
```

44. Weigh the evidence: Y carries more debt per dollar of equity and higher leverage, but also earns more cover for its interest bill.
45. Write the verdict: Y has the higher structural solvency risk (D/E 2.78, leverage 5.00) even though its interest coverage is currently stronger.

Test it

Your table matches the model answers (D/E 1.76 vs 2.78; leverage 3.82 vs 5.00; coverage 10.0 vs 12.7) and your verdict identifies Company Y as the riskier capital structure.

Note: The full worksheet for this activity is available on the LMS, and a printable copy is in activities/activity-06 of the course repository.

Revision Pointers for the Final Assessment

- K1/K2 — Be able to explain the statement of financial position: assets, liabilities, equity, and how the balance sheet equation stays in balance.
- K3 — Be able to trace the income statement from revenue to net income, and explain why profit is not cash (link to the cash flow statement).
- K4 — Know the statement of changes in equity: Beginning Equity + Net Income − Dividends ± Other changes = Ending Equity.
- K5 — Be able to pick the right analysis technique: ratio analysis, horizontal (trend) analysis, vertical analysis, benchmarking.
- A1 — Practise reading ratio trends across periods: what do a falling current ratio and a rising debt-to-equity together tell you?
- A2 — Practise the DuPont decomposition: ROE = Net Profit Margin × Asset Turnover × Equity Multiplier.
- Bring your completed activity worksheets — the assessment is open book.

Glossary

- **Balance sheet —** Statement of financial position: Assets = Liabilities + Equity, as at a date.
- **Income statement —** Profit & Loss statement: Profits = Revenues − Expenses, over a period.
- **Cash flow statement —** Cash movements in operating, investing and financing activities.
- **Statement of changes in equity —** Reconciles opening to closing equity for the period.
- **Working capital —** Current assets minus current liabilities.
- **EBITDA —** Earnings before interest, tax, depreciation and amortization.
- **EBIT —** Earnings before interest and tax — operating profit.
- **Current ratio —** Current assets / current liabilities — a liquidity measure.
- **Quick (acid test) ratio —** (Current assets − inventories) / current liabilities.
- **Debt-to-equity ratio —** Total liabilities (or long-term debt) / shareholders' equity.
- **Inventory turnover —** COGS / average inventories — stock efficiency.
- **ROE / ROA —** Return on equity / return on assets — profitability of capital.
- **Time value of money —** PV = FV / (1+i)^n — discounting future cash to today.
- **NPV —** Net present value: PV of inflows − initial outlay; accept if > 0.
- **Profitability index —** PV of inflows / PV of outflows; accept if > 1.
- **IRR —** The discount rate at which NPV equals zero.

- **Payback period —** Time for cumulative cash inflows to repay the initial outlay.
- **Variance —** Difference between budget and actual — adverse or favourable.
- **Horizontal analysis —** Percent change of the same line item across periods.
- **Vertical analysis —** Each line as a percent of a base figure (sales or total assets).
- **DuPont analysis —** ROE = Net Profit Margin × Asset Turnover × Equity Multiplier.